

Lecture 8 (10/2)

Correlation (2.4), Estimators, Bias, and Consistency

Standardized variables

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Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of Z_Y on Z_X ?

Standardized variables

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$$\begin{aligned} Z_Y &= (Y - \mu_Y) / \sigma_Y \\ &= [(\beta_0 + \beta_1 X + \epsilon) - (\beta_0 + \beta_1 \mu_X)] / \sigma_Y \quad (\text{substitute } Y, \mu_Y) \end{aligned}$$

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Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted ρ .

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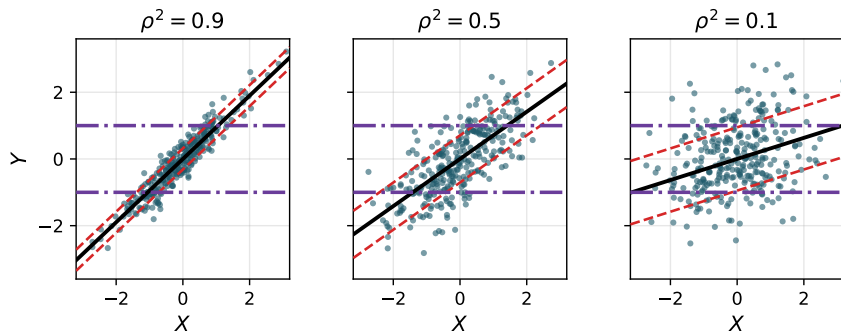
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$1 - \rho^2$ is the fraction of the variance in Y *left over* after conditioning on X .

Visualizing ρ^2



Here $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$ (hence $\text{cov}(X, Y) = \beta_1 = \rho$). The black lines show $E[Y | X] = \rho X$ and red lines $\pm \sigma_\epsilon = \pm \sqrt{1 - \rho^2}$. Purple: $\pm \sigma_Y = \pm 1$.

Summary: measures of association

	Formula	Units	Range
$\text{cov}(X, Y)$		units $Y \times$ units X	$(-\infty, \infty)$
β_1	$\frac{\text{cov}(X, Y)}{\sigma_X^2}$	units $Y/\text{units } X$	$(-\infty, \infty)$
ρ	$\frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$	none	$[-1, 1]$
ρ^2	$1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2}$	none	$[0, 1]$

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- $\hat{\theta}$ is random – it will vary depending on which sample we collect.
- θ is a fixed, non-random number (in frequentist statistics).

Sample distribution and standard error

Example

Let $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$ and estimate μ with $\hat{\mu} = \bar{Y} = \frac{1}{N} \sum_i Y_i$.

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$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

This is the sample distribution of $\hat{\mu}$ is its distribution over replications of the data. The standard deviation of the sample distribution is the standard error

$$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

Estimating q

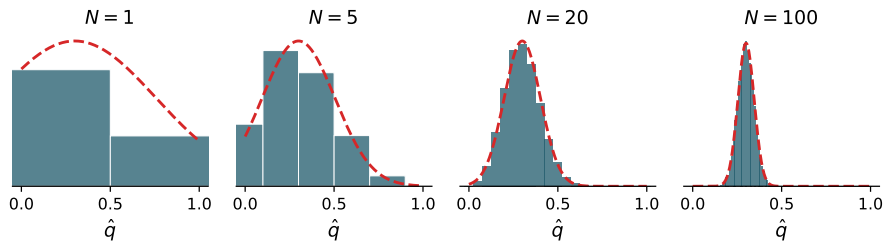
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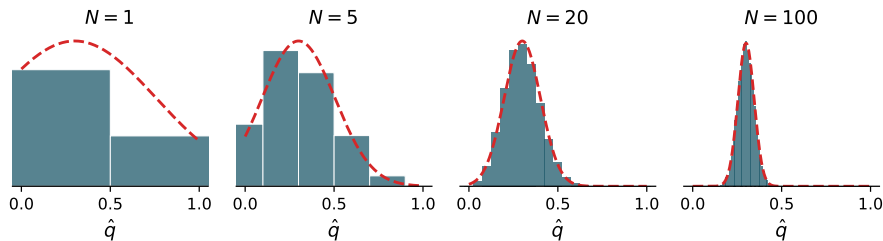
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$\hat{q}$ is not Normal, but by the CLT the Normal result carries over for large N , with $\sigma^2 = q(1 - q)$:

$$\hat{q} \approx \text{Normal} \left(q, \frac{q(1 - q)}{N} \right), \quad \text{se}(\hat{q}) = \sqrt{\frac{q(1 - q)}{N}}.$$

Bias and consistency

For any estimator, we define the mean squared error

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$\hat{\theta}$ is consistent if

$$E[\hat{\theta}] \rightarrow \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

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Example

$$\hat{\sigma}_0^2 = \frac{1}{n} \sum_i (Y_i - \bar{Y})^2$$

Is this unbiased?

Bessel's correction

Add and subtract μ , expand, and use $\sum_i (Y_i - \mu) = n(\bar{Y} - \mu)$:

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Taking expectations, $E[(Y_i - \mu)^2] = \sigma^2$ and $E[(\bar{Y} - \mu)^2] = \text{var}(\bar{Y}) = \sigma^2/n$:

$$E\left[\sum_i (Y_i - \bar{Y})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2.$$

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So $E[\hat{\sigma}_0^2] = \frac{n-1}{n}\sigma^2$ is biased and consistent. Dividing by $n-1$ (Bessel's correction) gives a unbiased estimator

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_i (Y_i - \bar{Y})^2, \quad E[\hat{\sigma}^2] = \sigma^2$$